

TrES-3b

R-0073

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250415 · created 2026-07-15T21:28:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460780.845 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.023
Transit depth (Rp/Rs)^2	3.79 +/- 0.89 [%]
Orbital Inclination (inc)	81.28 +/- 0.65
Airmass coefficient 1 (a1)	1.056 +/- 0.015
Airmass coefficient 2 (a2)	-0.0396 +/- 0.008
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#2 - [584, 399]
Optimal Method	PSF photometry
Transit Duration (day)	0.0527 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.195 ± 0.023 vs 0.16309 (deviation 1.39σ)
Duration fit vs published	0.0527 d vs 0.0567 d (deviation 0.53σ)
Mid-time O–C	-1.8 min
Depth SNR	8.5
Residual scatter	1.46 %

Light curve

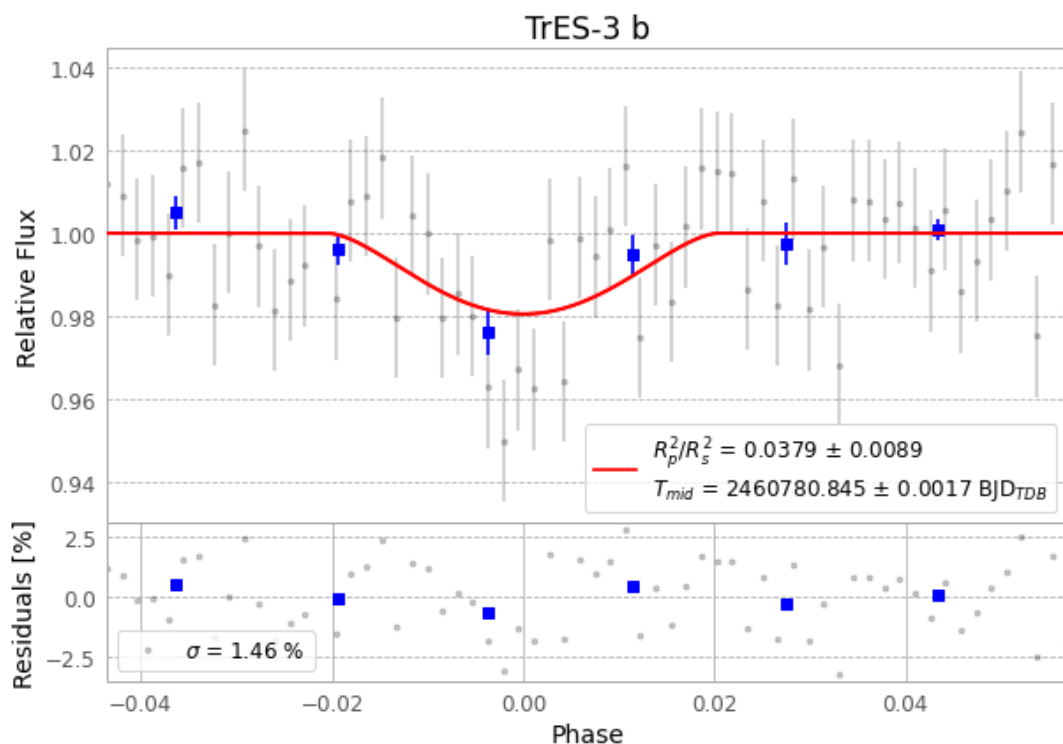


Figure 1 — FinalLightCurve_TrES-3 b_2025-04-14.png (SHA-256 b6d1d53d600c7d9e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0cdfb05efb9e473b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

64 FITS frames (42 MB), TRES-3250415065115.FITS → TRES-3250415100016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250415.log (e2884beb70eb...), FinalLightCurve_TrES-3 b_2025-04-14.png (b6d1d53d600c...), FinalLightCurve_TrES-3 b_2025-04-15.png (954e1bd76e97...), FinalLightCurve_TrES-3 b_2025-04-15.csv (294ffbbfa5ed...), FinalLightCurve_TrES-3 b_2025-04-14.csv (77a2aca4fcc5...)

Compute resources

Wall time 145.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:28Z.